

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

CELLULAR & MOBILE COMMUNICATIONS

(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) What is Cell splitting and explain? | 2M |
| (b) Write the differences between macro and micro cellular structures. | 2M |
| (c) Write the features of umbrella pattern antennas. | 2M |
| (d) Explain in detail about the importance of cell-site antennas. | 2M |
| (e) Mention any two advantages of Omni-directional antenna. | 2M |
| (f) Write short notes on umbrella pattern effect. | 2M |
| (g) Define paging channels. | 2M |
| (h) What is known as access channels? | 2M |
| (i) Define Handoff. | 2M |
| (j) Draw a simple two –level handoff scheme diagram. | 2M |

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

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|-----------|--|-----|
| 2 | Explain the concept of frequency reuse channels and frequency reuse distance. | 10M |
| OR | | |
| 3 | Describe about desired C/I from a normal case in an omni-directional antenna System. | 10M |
| 4 | Explain about the entire point to point model with its equations. | 10M |
| OR | | |
| 5 | What is the minimum separation required for cell site antennas and discuss about high gain antennas. | 10M |
| 6 | Explain the role of directional antennas for interference reduction of cellular systems. | 10M |
| OR | | |
| 7 | (a) Explain how co-channel interference is measured in real time mobile transreceiver. | 5M |
| | (b) Compare Frequency Diversity and Time Diversity Techniques. | 5M |
| 8 | Explain about channel sharing and channel borrowing concept in detail. | 10M |
| OR | | |
| 9 | Explain about cell sectoring technique. | 10M |
| 10 | Briefly explain measurement of averaged received signal level and level crossings. | 10M |
| OR | | |
| 11 | What is meant by handoff? Describe the classification of handoff processes. | 10M |
